



RESEARCHER BIAS: REPRODUCIBILITY CRISIS

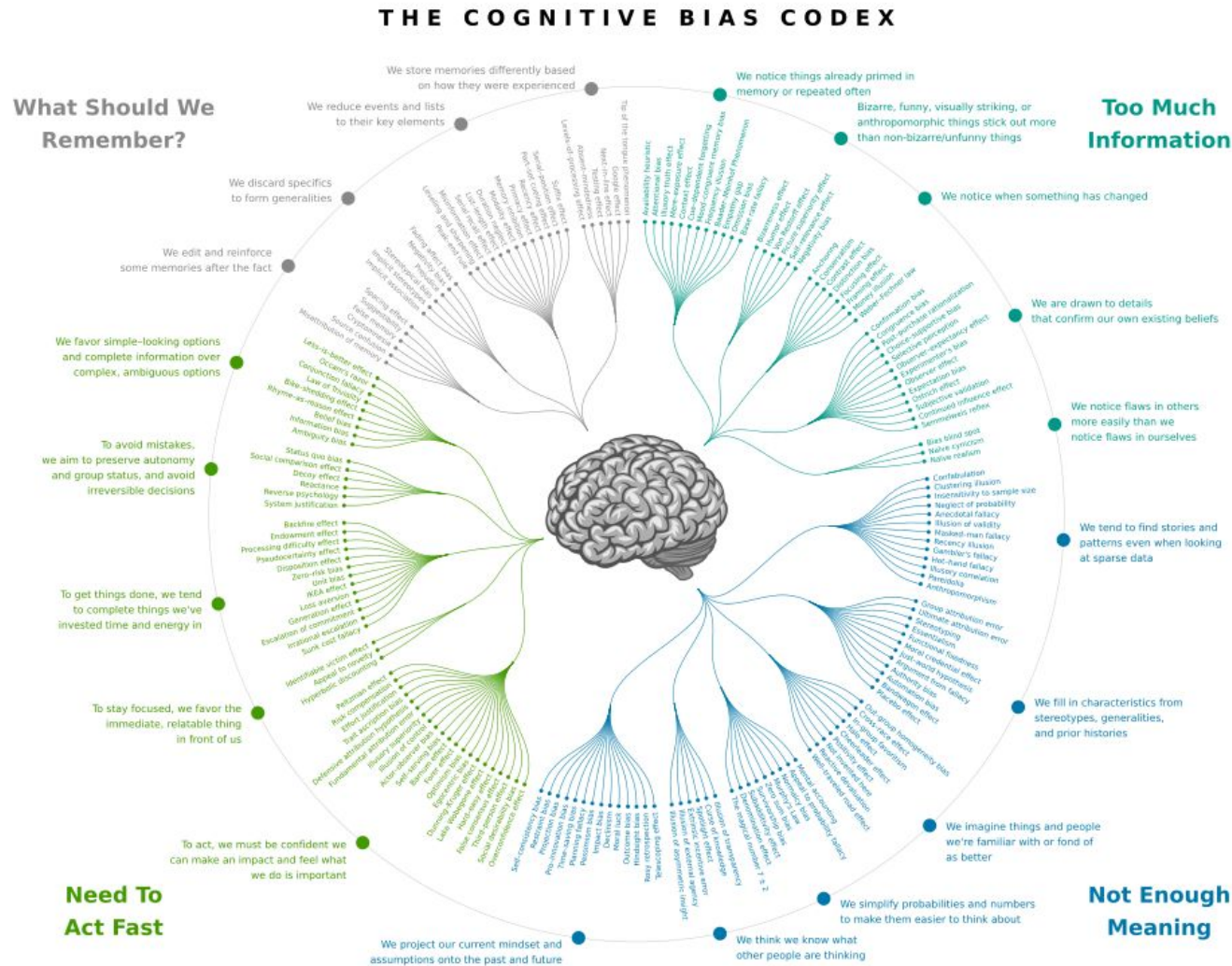
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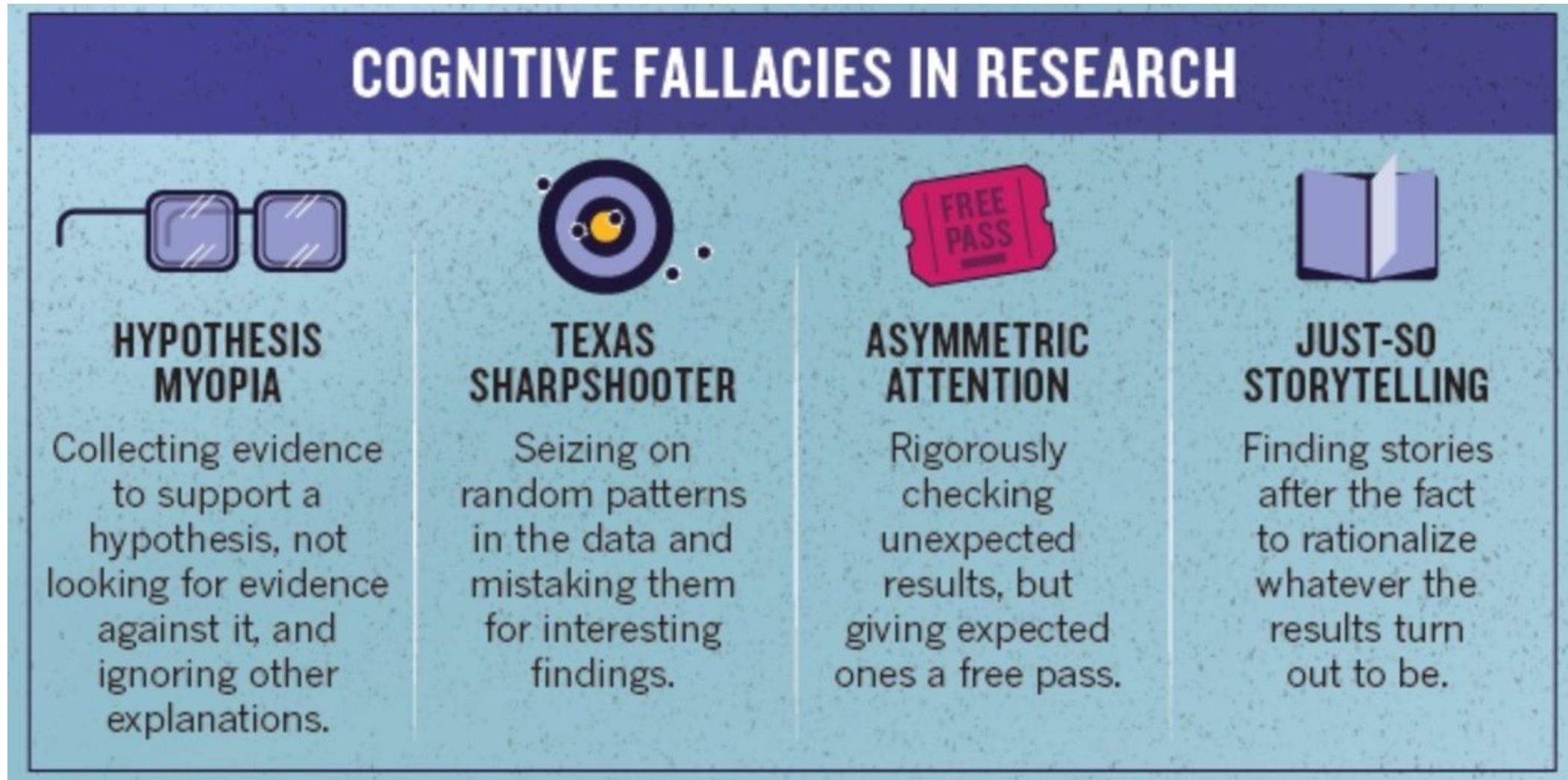


Cognitive biases affect researchers too



- Confirmation bias, Herding effect, Halo effect, ...
- Incentives to publish new discoveries discourage scientists from scrutinizing their work too much
- The very people who invest time and effort in confirmation of hypotheses are wrong people to try to make their hard-sought discovery disappear.
- Perverse incentives sometimes harm researchers

How researchers fool themselves



<https://www.nature.com/news/how-scientists-fool-themselves-and-how-they-can-stop-1.18517>



Hypothesis myopia

- Researchers tend to ask questions that give 'yes' answers if their favorite hypothesis is true
- They fixate on collecting evidence to support just one hypothesis
- Neglect to look for evidence against it
- Fail to consider other explanations.



The Texas sharpshooter

- A bad shooter who fires bullets at random at the side of a barn, then draws a “bull’s eye” around the biggest clump of bullet holes, and points proudly at his success.

The Texas Sharpshooter Procedure





p-hacking

- “p-hacking” - Exploiting “researcher degrees of freedom” until $p < 0.05$.
 - misuse of data analysis to find patterns in data that can be presented as statistically significant, thus understating the risk of false positives.
 - Risk: having a large number of features can lead to spurious but significant relationships
 - Underpowered studies with sample sizes too small to be reliable: Less powered a study, the more prone it is to find a result that isn't real.
 - Another bad practice: Researchers stop collecting data when results reach statistical significance. (Compare to: getting three heads in a row when flipping a coin, and then concluding that coin flips always end on heads.)

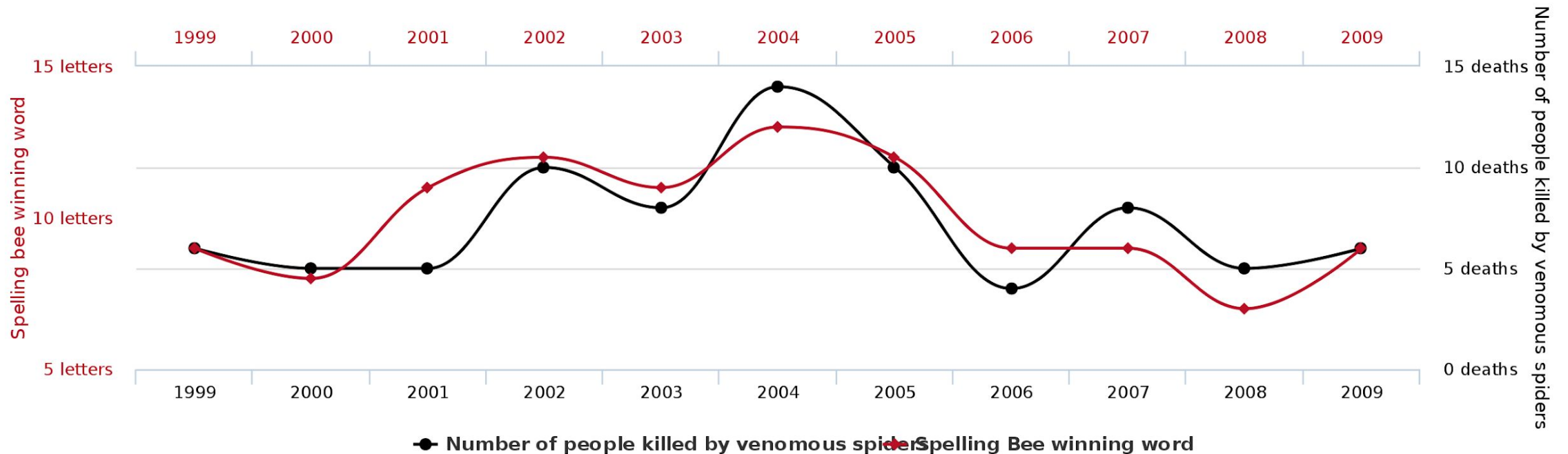


Spurious correlations

Letters in winning word of Scripps National Spelling Bee

correlates with

Number of people killed by venomous spiders



tylervigen.com



Bonferroni correction

- statistical method used to counteract the problem of multiple comparisons, i.e., testing several hypotheses at once, which raises the chance of a false positive.
- E.g., effect is statistically significant at α level if the probability of observing it by chance p is smaller than α
- If you are testing m hypotheses for statistical significance at level α (e.g., $\alpha = 0.05$),
- The Bonferroni correction tests each hypothesis at α/m . (i.e. $p < \alpha/m$)



Just-so storytelling

- post-hoc stories can be concocted to justify anything and everything
- “HARK-ing” - hypothesizing after results are known

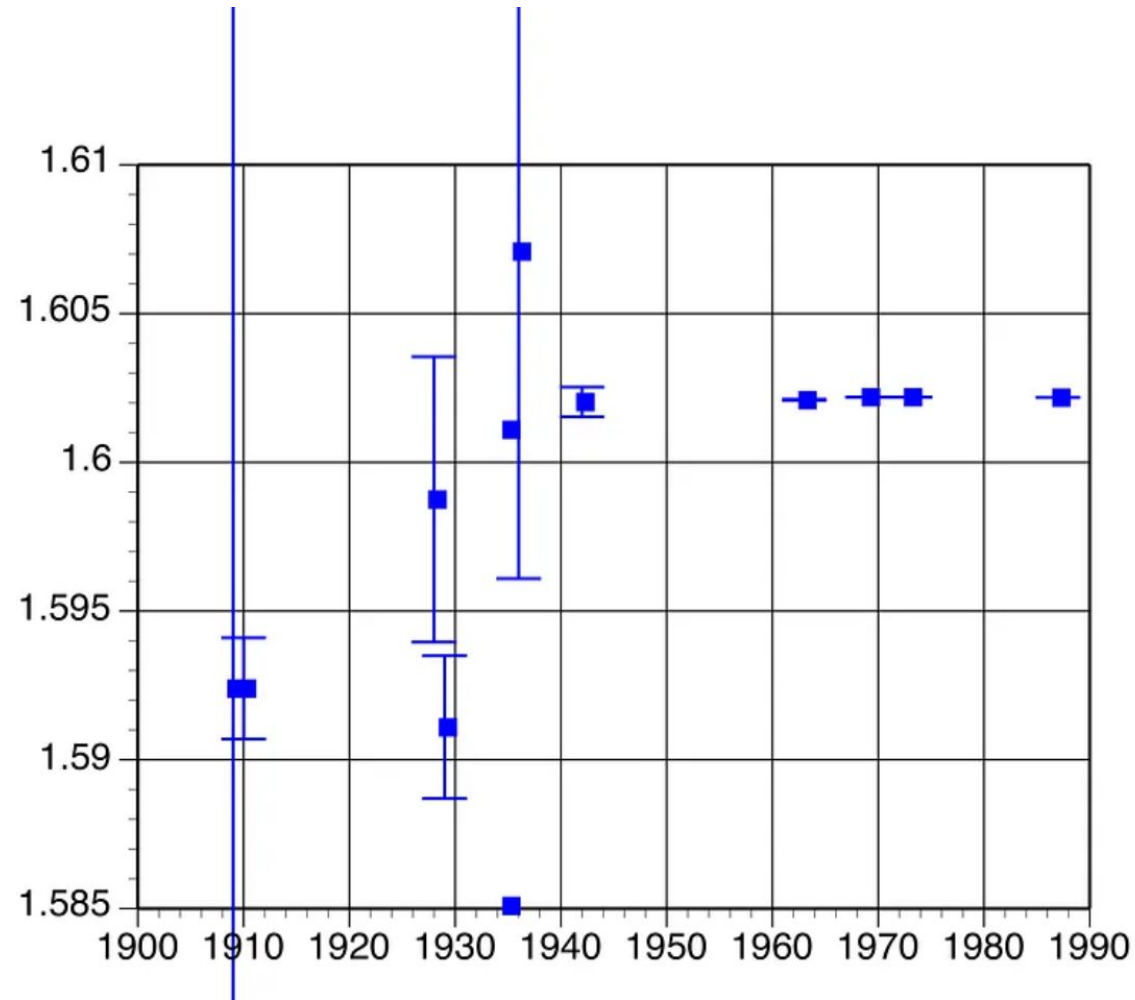


Asymmetric attention

- **Confirmation bias:** Tendency to interpret evidence in ways that support prior beliefs
- Leads to asymmetric attention, aka disconfirmation bias - when researchers accept the results they *expected*, but we rigorously check *unexpected* results
- A study of 165 laboratory experiments found that in 88% of cases in which results did not align with expectations, the scientists blamed the inconsistencies on how the experiments were conducted, rather than on their own theories. Consistent results, by contrast, were given little to no scrutiny.



Milliken experiment to measure electron charge (1908)



Millikan (notebooks)
Millikan (published)
Erik Backlin, Nature 1929
[Birge, 1929]
Backlin and Flemberg, Nature 1936
Backlin and Flemberg, cited in HR Robinson RPP 1937
Gunnar Kellström PR 1936
[Birge, 1942]
[Dummond and Cohen, 1963]
[Taylor et al, 1969]
[Cohen and Taylor, 1973]
[Cohen and Taylor, 1987]



Sampling bias

- Non-representative samples skew results
 - E.g., Using WEIRD (Western, Educated, Industrialized, Rich and Democratic) participants for generalizable claims
 - Most psychology experiments are conducted on US undergraduates



Publication bias (file-drawer effect)

- Journals tend to publish papers with positive conclusions.
 - A scientist runs two experiments: One works, one doesn't (negative results).
 - The one that works is submitted to the journal
 - The one that doesn't stays in a drawer, aka “the file drawer effect”.
- This changes the population of experiments, making it difficult to systematically study effects (e.g., through meta review)



Why Most Published Research Findings Are False

John P. A. Ioannidis

Summary

There is increasing concern that most current published research findings are false. The probability that a research claim is true may depend on study power and bias, the number of other studies on the same question, and, importantly, the ratio of true to no relationships among the relationships probed in each scientific field. In this framework, a research finding is less likely to be true when the studies conducted in a field are smaller; when effect sizes are smaller; when there is a greater number and lesser preselection of tested relationships; where there is greater flexibility in designs, definitions, outcomes, and analytical modes; when

factors that influence this problem and some corollaries thereof.

Modeling the Framework for False Positive Findings

Several methodologists have pointed out [9–11] that the high rate of nonreplication (lack of confirmation) of research discoveries is a consequence of the convenient, yet ill-founded strategy of claiming conclusive research findings solely on the basis of a single study assessed by formal statistical significance, typically for a p -value less than 0.05. Research is not most appropriately represented and summarized by p -values, but, unfortunately, there is a widespread notion that medical research articles

is characteristic of the field and can vary a lot depending on whether the field targets highly likely relationships or searches for only one or a few true relationships among thousands and millions of hypotheses that may be postulated. Let us also consider, for computational simplicity, circumscribed fields where either there is only one true relationship (among many that can be hypothesized) or the power is similar to find any of the several existing true relationships. The pre-study probability of a relationship being true is $R/(R + 1)$. The probability of a study finding a true relationship reflects the power $1 - \beta$ (one minus the Type II error rate). The probability of claiming a relationship when none

CRISIS OF REPRODUCIBILITY



What is reproducibility & replicability

- Independent verification is a cornerstone of science. Experiment needs to be replicated to find out if the measured effects are consistent
- Reproducibility is the ability of an independent researcher to obtain the same results using the original data and methods.
 - Reproducibility: same data, same methods, same results
 - Replicability: new data, same methods, similar results
 - Robustness: same data, different methods, similar conclusions
 - Generalizability: new data, same methods, similar results



The “Crisis” of reproducibility in science

- But, across many disciplines—especially psychology, medicine, and social sciences—a large number of published findings cannot be reliably reproduced or replicated.
 - 60% of 100 psychology studies could not be replicated
 - Results of only 13 of 21 studies published in *Nature* and *Science* replicated.
- Why?
 - Rarely, fraud
 - Every time scientists conduct an experiment, there's a chance they'll find a false positive (esp. if p-hacking)
 - Underpowered studies that produce significant results from small samples are often statistical flukes
 - Participant population may be different across studies



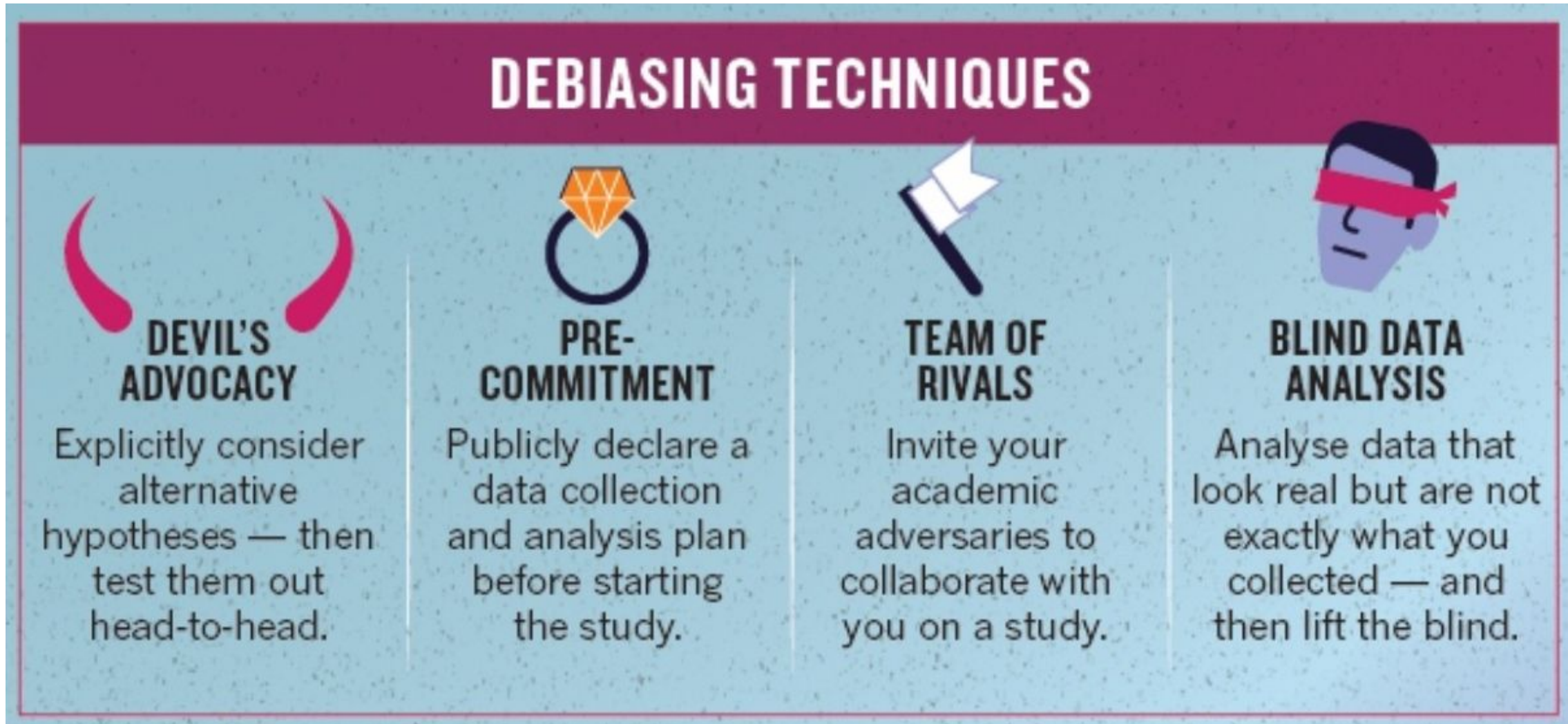
Some well-known non-replicable ideas

- Many underpowered non-reproducible studies made its way into pop culture
 - **Power poses:** In 2010 an experiment with 42 participants showed that people feel more powerful if they posed with an open, expansive posture.
 - **Marshmallow test:** ability to delay gratification in childhood is associated with adult success.
 - **Ego depletion:** will power is a finite resource
 - **Social priming:** behavior can be nudge by planting ideas
 - **Google memory effect:** people don't remember as well when they know they can look things up



DEBIASING SCIENCE

Debiasing research





Strong inference

- explicitly consider competing hypotheses,
- develop experiments that can distinguish between them



Transparency

- Open science: researchers share their methods, data, computer code and results in central repositories
- Open Science Framework
 - Hypothesis pre-registration: Precommit to analysis and report plan to mitigate the effect of cognitive biases
 - Registered report: publication in which scientists present their research plans for peer review before they even do the experiment.
 - Also avoids publication bias



Teams of rivals

- a good debiasing approach is to bypass the typical academic back-and-forth and instead invite your academic rivals to work with you
- rivals will quickly spot flaws such as hypothesis myopia, asymmetric attention or just-so storytelling, and cancel them out with similar slants favouring the other side.



Blind data analysis

- Researchers who do not know how close they are to desired results will be less likely to find what they are unconsciously looking for
- Data blinding approaches
 - Create alternative data sets, e.g., adding random noise or a hidden offset to data
 - Move participants to different experimental groups
 - Hiding demographic categories of data
- Researchers analyze fake data set as usual — cleaning the data, handling outliers, running analyses
- In the background, a computer applies all of their actions to the real data.



An illustration “team of rivals”

[Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results, Silberzahn et al.]

- Are scientific results are highly contingent on subjective decisions at the analysis stage?
- 29 teams analyzed the same data to answer same research question.

Do darker skinned soccer players get more red cards?

25



- Red card results in the player's ejection from the game and his team must play with one fewer player for the remainder of the match (big disadvantage)
- Red cards are given for aggressive behavior (tackling, intentionally fouling the opponent a goal-scoring opportunity, hitting, or using threatening and abusive language).
- But, the evidence is often ambiguous, and referee decision may be biased



Data

- All teams were given the same large data set collected by a sports-statistics firm across four major football leagues, games, years, referees, players.
- Referee calls, counts of how often referees encountered each player, and player demographics including team position, height and weight.
- Includes a rating of players' skin color, coded manually:
 - two independent coders sorted photographs of players into five categories ranging from 'very light' to 'very dark' skin tone.



Some considerations of analysis

- Do you treat each red-card decision as an independent observation?
- How do you account for some referees giving more red cards than others?
- Would you take into account whether a referee's familiarity with a player affects the referee's likelihood of assigning a red card?
- Would you look at whether players in some leagues are more likely to receive red cards compared with players in other leagues, and whether the proportion of players with dark skin varies across leagues and player positions?



Project Stage	Work Package	Month																			
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	Building the Data Set																				
2	Recruitment and Initial Survey of Data Analysts																				
3	First Round of Data Analysis																				
4	Round-Robin Peer Evaluations																				
5	Second Round of Data Analysis																				
6a	Open Discussion and Debate, Further Analyses																				
6b	Write-Up of Manuscript																				
7	Internal Experts' Peer Review of Approaches																				
	Revision of Manuscript																				



Some data statistics

Height (cm)	$M = 181.74$ ($SD = 6.69$)
Weight (kg)	$M = 75.64$ ($SD = 7.10$)
Number of games	$M = 71.13$ ($SD = 36.17$)
Number of yellow cards	$M = 27.41$ ($SD = 24.08$)
Number of red cards	$M = 0.89$ ($SD = 1.26$)
League country	
England	$n = 564$ players
France	$n = 533$ players
Germany	$n = 489$ players
Spain	$n = 467$ players
Skin color	
0 (very light skin)	Rater 1: $n = 626$ players Rater 2: $n = 451$ players
.25	Rater 1: $n = 551$ players Rater 2: $n = 693$ players
.50	Rater 1: $n = 170$ players Rater 2: $n = 174$ players
.75	Rater 1: $n = 140$ players Rater 2: $n = 141$ players
1 (very dark skin)	Rater 1: $n = 98$ players Rater 2: $n = 126$ players



Some analytic approaches

Team	Distribution	Treatment of nonindependence	Number of covariates	Analytic approach
1	Linear	Clustered standard errors	7	Ordinary least squares regression with robust standard errors, logistic regression
6	Linear	Clustered standard errors	6	Linear probability model
14	Linear	Clustered standard errors	6	Weighted least squares regression with clustered standard errors
4	Linear	None	3	Spearman correlation
11	Linear	None	4	Multiple linear regression
10	Linear	Variance component	3	Multilevel regression and logistic regression
2	Logistic	Clustered standard errors	6	Linear probability model, logistic regression
30	Logistic	Clustered standard errors	3	Clustered robust binomial logistic regression
31	Logistic	Clustered standard errors	6	Logistic regression
32	Logistic	Clustered standard errors	1	Generalized linear models for binary data



Covariates used by teams

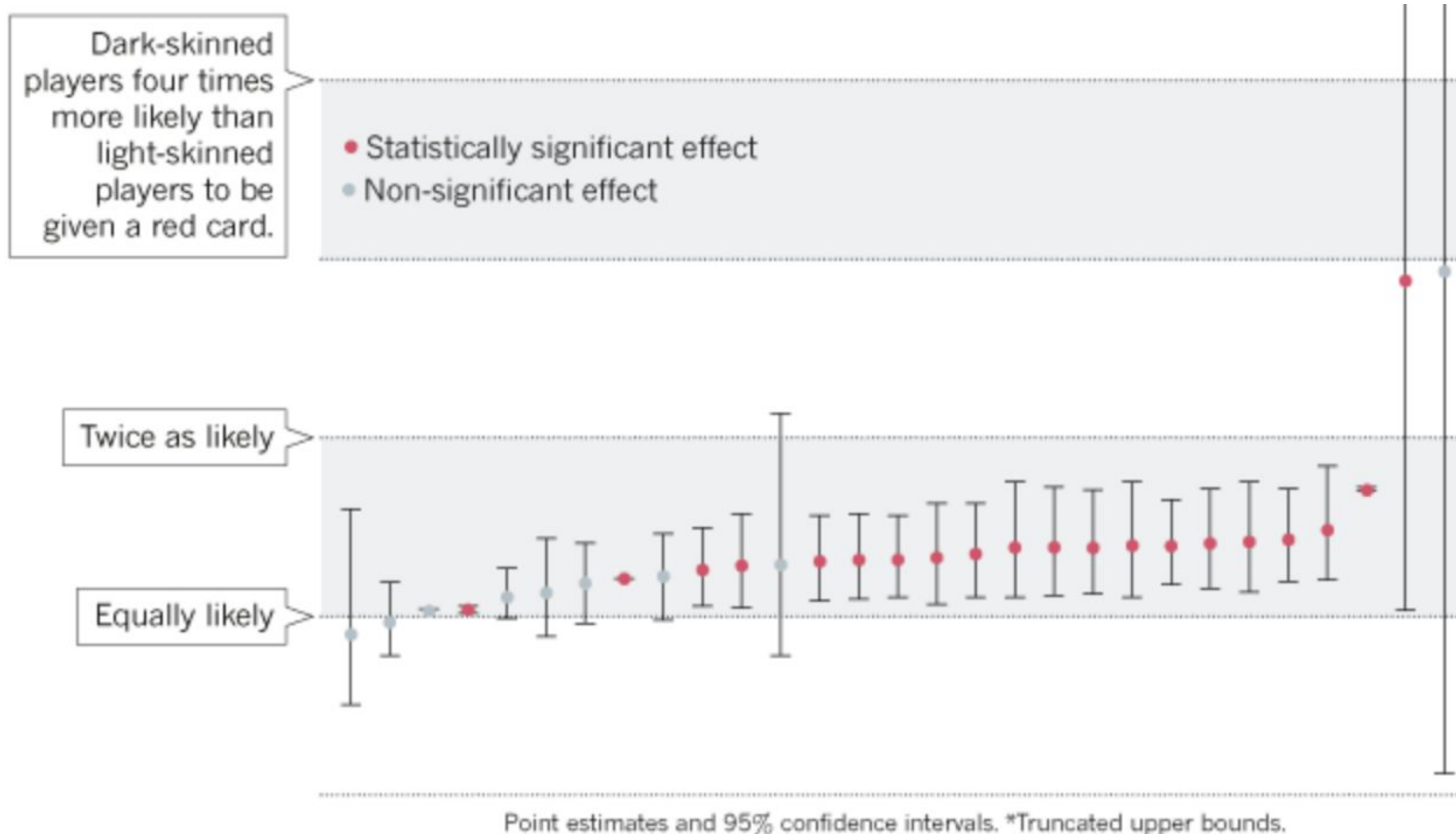
Covariate	1	2	3	4	5	6	...
Player position	X	X	X			X	
Player's height	X	X		X		X	
Player's weight	X	X		X		X	
Player's league country ^a	X						
Player's age	X					X	
Goals scored by player		X					
Player's club	X					X	
Referee's country		X	X	X			
Referee	X					X	
Player's number of victories		X					
Number of cards received by player							
Player							
Number of cards awarded by referee							
Number of draws							
Number of covariates	7	6	2	3	0	6	



Analytic Approach	Odds Rat
Zero-Inflated Poisson Regression	0.89
Bayesian Logistic Regression	0.96
Hierarchical Log-Linear Modeling	1.02
Multilevel Regression and Logistic Regression	1.03
Hierarchical Bayes Model	1.10
Logistic Regression	1.12
OLS Regression With Robust Standard Errors, Logistic Regression	1.18
Spearman Correlation	1.21
WLS Regression With Clustered Standard Errors	1.21
Multiple Linear Regression	1.25
Clustered Robust Binomial Logistic Regression	1.28
Linear Probability Model	1.28
Hierarchical Generalized Linear Modeling With Poisson Sampling	1.30
Multilevel Logistic Regression Using Bayesian Inference	1.31
Mixed-Model Logistic Regression	1.31
Hierarchical Poisson Regression	1.32
Linear Probability Model, Logistic Regression	1.34
Generalized Linear Mixed Models	1.38
Multilevel Logistic Regression	1.38
Mixed-Effects Logistic Regression	1.38
Generalized Linear Models for Binary Data	1.39
Negative Binomial Regression With a Log Link	1.39
Cross-Classified Multilevel Negative Binomial Model	1.40
Poisson Multilevel Modeling	1.41
Multilevel Logistic Binomial Regression	1.42
Generalized Linear Mixed-Effects Models With a Logit Link	1.48
Dirichlet-Process Bayesian Clustering	1.71
Tobit Regression	2.88
Poisson Regression	2.93



Odds ratio of red-carding a darker player



Can social outcomes be predicted?



- Matthew Salganik, Ian Lundberg, Alex Kindel, Sara McLanahan, et al
- Mass collaboration involving 457 researchers.



Birth to age 9
12,942 variables

4,242 families

Information about child and family



Given:

Birth to age 9
12,942 variables

4,242 families

Information about child and family

Background data

Predict:

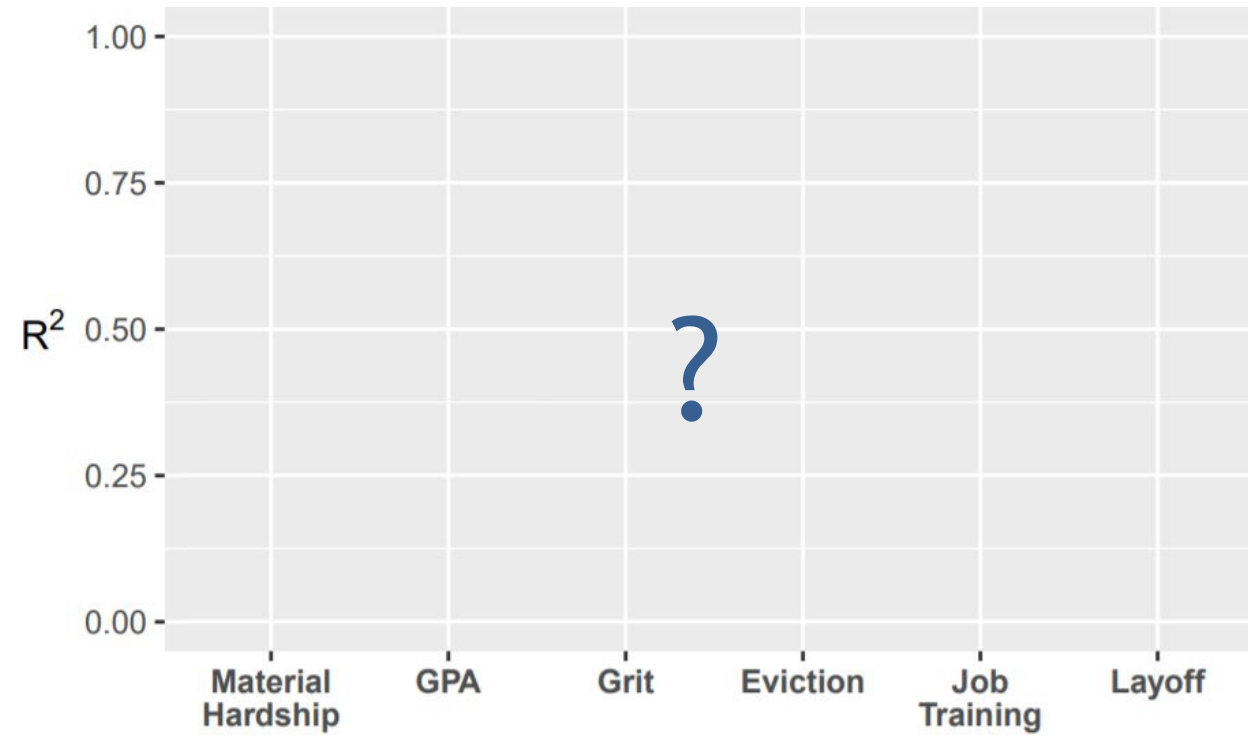
Age 15
6 variables

Training

Leaderboard

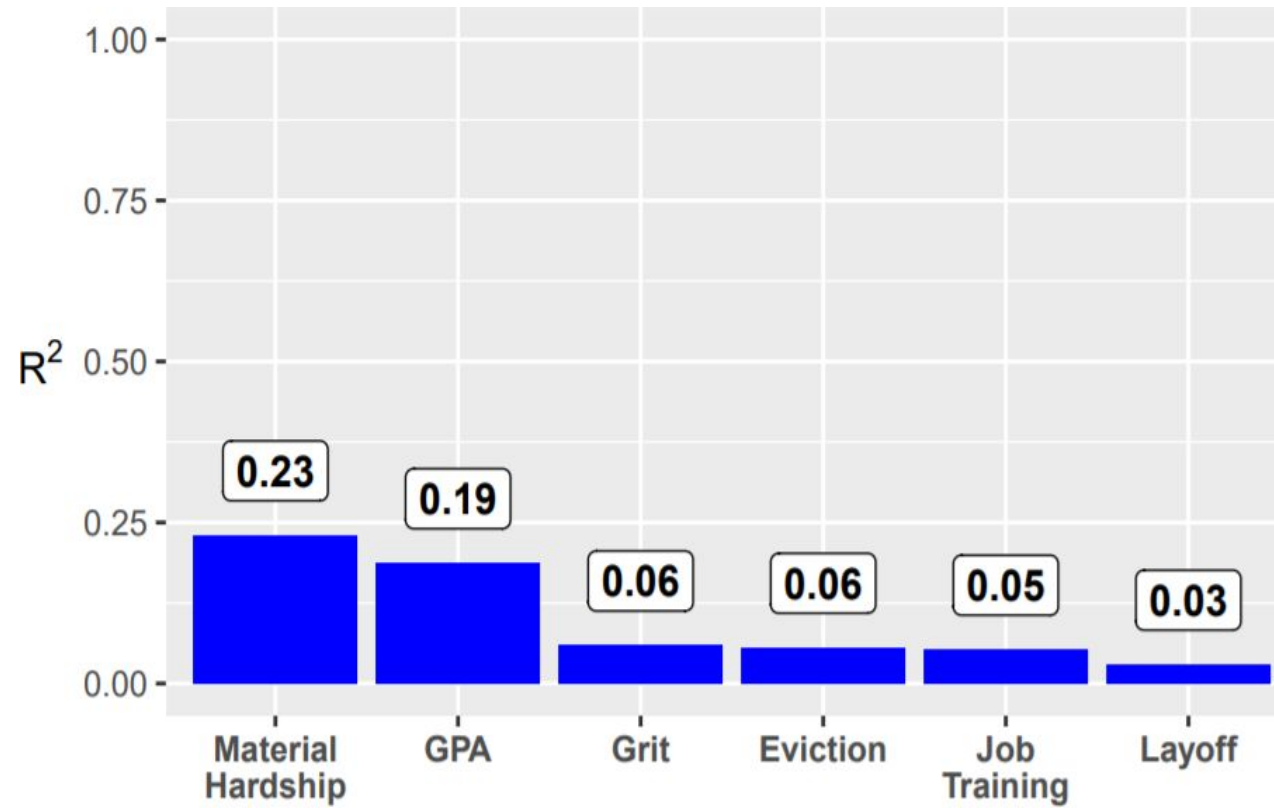
Holdout

Outcome
data



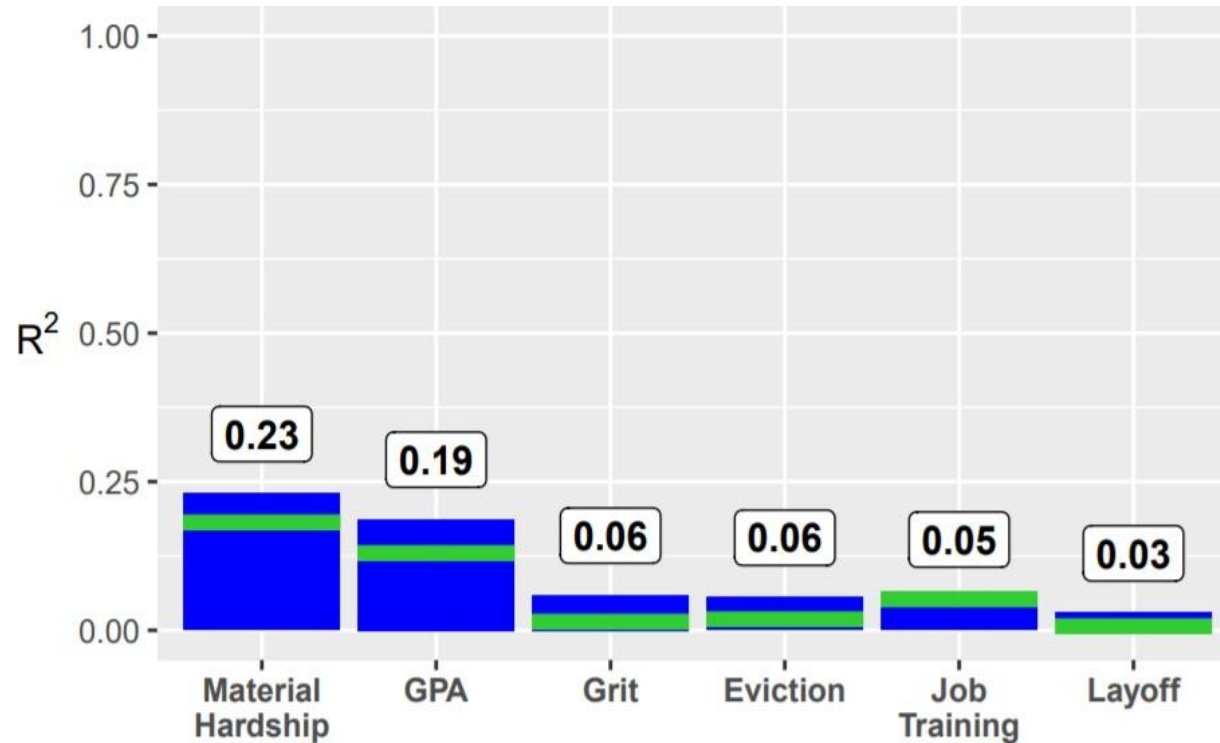


Best performing models





Green line: 4-variable linear regression



- 13,000 features hardly better than 4 features!
- “AI” hardly better than simple linear formula



Accuracy of recidivism prediction

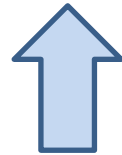
COMPAS tool (137 features):

65% $\pm$ 1%

Logistic regression (2 features):

67% $\pm$ 2%

(slightly better
than random)



Age and number of priors

[Dressel & Farid. *The accuracy, fairness, and limits of predicting recidivism*. Science Advances 2018]



THE CREEP OF BIAS: BMI ILLUSTRATION

[CREDIT: [MAINTENANCE PHASE](#)]



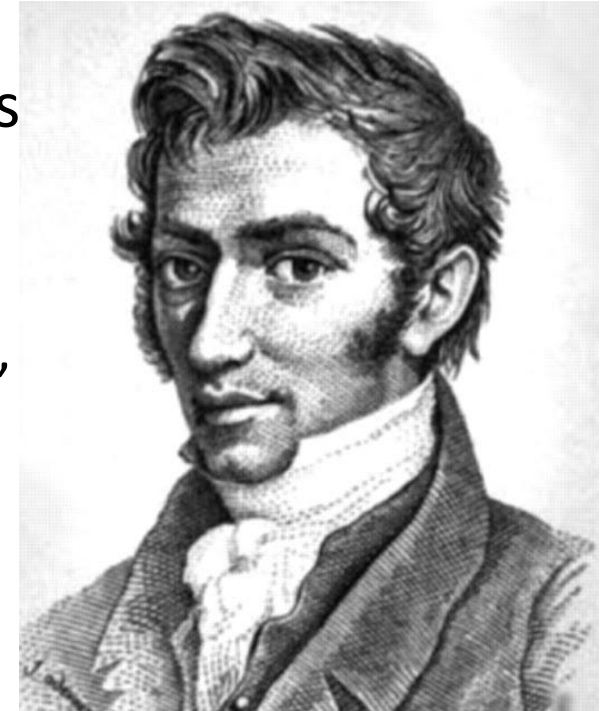
BMI

- **Body Mass Index (BMI)** is a person's weight in kilograms (or pounds) divided by the square of height in meters (or feet).
- Has become central to accessing health care
- Reduces a complex phenomenon (health) to a single measure
- BMI is a proxy of body fatness.
- Ubiquitous and discriminatory
- Shapes our understanding of health, research questions about health and how the data is collected for medical research



History of the BMI

- Adolphe Quetelet: Statistician doing population analysis
- Creator of “Social physics”: laws of human behavior
 - Wants to measure courage, heroism
 - Ends up analyzing state data, looking for patterns: birth rates, death rate, etc.
 - The “average man”
- Quetelet’s index (1842)
 - Ratio of weight and height has a normal distribution
 - Height and weight data from France and Scotland (only white men included)
 - Population level description (warns against using it on individuals)

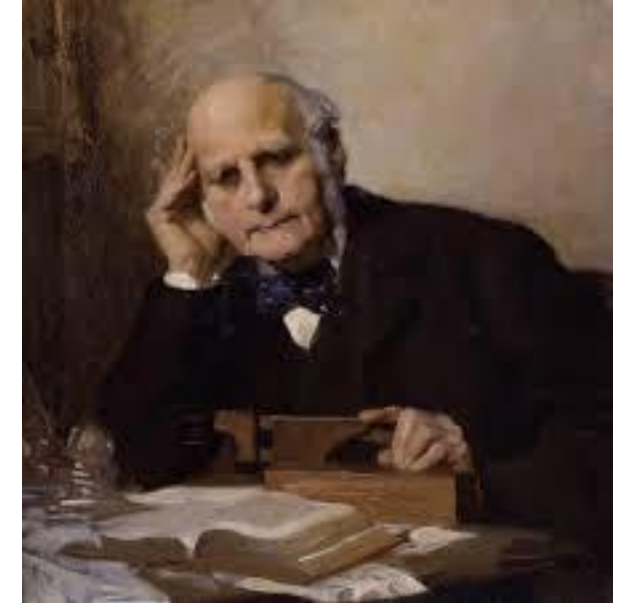


Adolphe Quetelet
b. 1796 Belgium



History of the BMI

- Eugenics movement picks up on the idea
- Builds up on Quetelet's ideas, but instead of equating the "normal" as ideal and desirable, Galton considers normal to be mediocre, something to be overcome



Francis Galton



History of the BMI

- American life insurance company creates height/weight tables to charge fair policy holders more (1867)
- Metlife insurance – the first aligned insurance table (1942)
 - Based on data (height and weight measurements) of policy holders
- Doctors start using “normal” body size in individual health recommendations



As a medical diagnostic tool

- 1972 Ancel Keys, “Indices of Relative Weight and Obesity”
- Study on body-fat percentage:
 - 7400 men from five countries
 - Compared formulas to directly measured values (water displacement, skin fold test)
- Best predictor
 - Quetelet’s formula for the “average man”: $\text{weight}/\text{height}^2$
 - Keys named it “Body Mass Index”
 - But, only explains 50% of the variance of fatness



BMI becomes a medical standard

- BMI becomes the basis for standardizing bodies (1990s)
 - Categories of overweight, obese, etc. being used as a predictor of health risk
- 1995 WHO decides BMI becomes a global measure of fat and obesity
 - International health policy priority
 - Despite mounting evidence that people can be healthy at higher weights
 - NIH adopts these guidelines in 1998
 - Changing standards creates an “epidemic” of obesity
 - “Millions of Americans became fat even if they did not gain a pound”
 - Medical community starts to “backfill the science” with studies about dangers of obesity
 - WHO sponsors Roche and Abbott had anti-obesity drugs approved shortly after
 - Codify the arbitrary scale



Problems of using BMI

- Ecological fallacy of BMI
- Does BMI clarify or muddy our understanding of weight and health?
 - BMI ignores many different types of obesity
 - BMI ignores racial differences
 - Correlation between BMI and disease shows up at lower/higher BMI depending on race
 - Pacific Islanders were historically healthier at larger BMI
 - Some countries with lower BMI have highest Type 2 diabetes rates
 - BMI ignores gender
 - Missing opportunities to intervene on public health issues
- BMI is poor proxy of poor health
 - At population level, higher BMI is associated with worse health outcomes
 - But other measures (exercise) are better correlated with health outcomes
 - Direct measures (resting HR, cholesterol), lifestyle (exercise) are better predictors of health risk
 - We try to control the proxy of health risk (BMI) rather than directly controlling the health risk (through lifestyle interventions)



BMI and health: the jury is still out

- Population-level correlation between higher BMI and poor health outcomes
 - 1/3 of “obese” people have normal health markers, are healthy
- Medical bias as a driver of “health effects of obesity”
 - Life threatening illnesses are often not diagnosed
 - Doctors refuse to run tests or treat patients until they lose weight
 - People postpone care for fear of being stigmatized and judged
 - “HONDA” – terms used by doctors to describe patients (hypertensive, obese, non-compliant, diabetic, alcoholic)
- Needed interventions to decrease bias among medical providers, as attitudes drive patient treatment
- Obesity paradox
 - Fat protects some people from some health conditions



Protecting yourself against blind spot bias

- All people are susceptible to bias (while everyone believes they are less-biased than others)
- Practice good mental habits to reduce bias
 - Be curious (vs avoid the unfamiliar)
 - Question everything
 - Diversify your social circles
 - Nurture your brain (hungry, tired brain is prone to quick, biased decisions)
 - Allow that your opinions will change